

Master of Science in Computer Science

Name of Programme	Master of Science in Computer Science
Programme Overview	Aspiring to excel in the tech industry? Our Master of Science in Computer Science is an advanced, flexible programme designed to strengthen both technical expertise and analytical capability. It covers core computing principles alongside advanced topics such as software development, data analysis, artificial intelligence, and cybersecurity. Students gain practical experience with industry-standard tools, mastering algorithms, software engineering, and secure system design. The programme builds strong critical thinking and problem-solving skills, preparing graduates to create scalable solutions and address ethical challenges. Career opportunities include roles in software engineering, data science, and cybersecurity, with the potential to drive innovation in the digital era.
Admission Requirements	Bachelors or Honours Bachelors Degree from a recognised institution
Technical Requirements	To ensure a smooth and effective learning experience, students enrolling in this programme must meet the following technical requirements: Device Requirements Computer/Laptop: Windows (10 or later) or Mac (macOS 10.15 or later); Processor: Intel i3 (or equivalent) or above; RAM: Minimum 4 GB (8 GB recommended for multitasking); Storage: Minimum 20 GB free space for coursework and downloads; Camera and Microphone: Built-in or external for live sessions and presentations OR Tablet/Smartphone: Suitable for accessing content on the go (limited functionality for assignments/exams) Internet Requirements Connection speed: Minimum 5 Mbps download and 2 Mbps upload (10 Mbps recommended for seamless video conferencing); Stable Wi-Fi or Ethernet connection recommended for live Zoom sessions Software and Platforms Virtual Live Sessions: Zoom (synchronous learning); Document Processing: Microsoft Office (Word, Excel, PowerPoint) or equivalent (Google Docs) Browser Requirements Supported browsers: Chrome (latest version), Safari (for Mac users); Cookies and JavaScript: Enabled for full Blackboard functionality Additional Tools (Optional but Recommended) Headset: For clear audio during virtual classes; External storage (USB/Cloud): For backups of important coursework; Antivirus software: To protect against malware or data loss
Typical Full-Time Study Period	2 years
Number of Modules	9 (7 core + 2 elective)
Total Programmes Credits	240
Fees per Programme Credit	\$19.50
Total Tuition Fee	\$4680

SEMESTER 1

Algorithms and Data Structures

Code: C9-ADS-25	This module enhances students' problem-solving abilities by introducing key algorithms and data structures. It empowers students to solve simple problems, apply advanced techniques, and improve their programming skills. Students will evaluate algorithms, analyse time and space complexities, and select appropriate data structures to design solutions for various problems.
Module Fees: \$312	
Module Credits: 16	

Object-Oriented Software Engineering

Code: C9-OSE-25	This module equips students with the skills to manage software projects and develop systems using object-oriented methods. Students will learn key concepts and techniques for system design and program implementation using rapid application development tools and designing graphical user interfaces to create, test, and implement software solutions.
Module Fees: \$312	
Module Credits: 16	

Operating Systems

Code: C9-OPS-25	This module reviews the design, implementation, and functioning of operating systems. It explores the principles behind centralised and distributed systems, focusing on real-world applications. Students will understand the structure and functions of operating systems, analysing requirements for specific scenarios and relating system functions to machine architecture.
Module Fees: \$312	
Module Credits: 16	

SEMESTER 1 ELECTIVE GROUP (CHOOSE ONE)

Machine Learning

Code: C9-MLE-25	This module provides an in-depth introduction to supervised and unsupervised machine learning, covering key models and algorithms for regression, classification, clustering, and probabilistic classification. Students will critically evaluate approaches, understand applications, design machine learning systems, apply problem-solving techniques, and develop skills for further research in the field.
Module Fees: \$234	
Module Credits: 12	

Web Engineering

Code: C9-WEN-25	This module provides hands-on experience in designing and developing internet-based applications. Students will learn internet architecture, HTML, CSS, JavaScript, server-side development, XML concepts, and web services. By completion, they will gain practical skills in building web applications and writing programs for front-end and back-end development
Module Fees: \$234	
Module Credits: 12	

SEMESTER 2

Data Warehousing

Code: C9-DWH-25	This module covers key concepts and components of data warehousing techniques and applications. Students will gain in-depth knowledge of data warehousing, including design methodologies, OLAP techniques, and development. They will design, implement, and evaluate data warehousing applications, analysing and applying core concepts for effective data management and analysis.
Module Fees: \$312	
Module Credits: 16	

Computing Research Methods

Code: C9-CRM-25	Computing Research Methods explore systematic approaches to investigating computational problems, developing algorithms, and evaluating systems. It includes qualitative and quantitative techniques, experimental design, data analysis, and simulation. These methods ensure rigour in software engineering, AI, cybersecurity, and more, driving innovation and advancing knowledge in computer science and information technology.
Module Fees: \$312	
Module Credits: 16	

Computer Networking and Communications

Code: C9-CNC-25	This module introduces the principles of structured networks and future challenges for network designers. It covers network protocols, internet technologies, and data communications. Students will explore familiar concepts like the web and email before studying more complex networking aspects, including LAN, WAN, MAN, and network security.
Module Fees: \$312	
Module Credits: 16	

SEMESTER 2 ELECTIVE GROUP (CHOOSE ONE)

Grid Computing Systems

Code: C9-GCS-25	This course explores advanced distributed computing, focusing on service-oriented systems like Grid Computing. Students will learn technologies, architectures, and applications, including cluster and data grids. It covers IT infrastructure, business value, middleware, and grid-enabled services, equipping students to design, develop, and implement grid computing models and applications.
Module Fees: \$234	
Module Credits: 12	

Information Retrieval

Code: C9-INR-25	This module introduces students to information retrieval systems, focusing on retrieving the best possible information for user queries. It covers key algorithms, evaluation methods, probabilistic retrieval, ranking principles, and real-world applications in web search and recommendation systems, along with emerging topics like MapReduce, portfolio retrieval, and online advertising.
Module Fees: \$234	
Module Credits: 12	

SEMESTER 3 (ONE YEAR)

Supervised Research and Dissertation

Code: C9-SRD-25	This module involves independent, in-depth study under expert guidance, culminating in a scholarly thesis. It includes topic selection, literature review, methodology development, data analysis, and findings presentation. This process hones critical thinking, research skills, and subject expertise, contributing original knowledge to the academic and professional fields.
Prerequisite: C9-CRM-25	
Module Fees: \$2340	
Module Credits: 120	

Globally Recognized
proudly African

online.bothouniversity.com

Disclaimer : Unauthorized copying or reproduction of this content is strictly prohibited